

Konstantinos Bizanos

Ph.D. Student in Mathematics

University of Missouri

kbf66@missouri.edu | kostasbizanos@gmail.com

<https://bizanosk.github.io/>

1 Education

- i. **University of Missouri, USA** Aug 2025 – Present
Ph.D. in Mathematics
- ii. **National and Kapodistrian University of Athens (NKUA), Greece** Oct 2022 – Feb 2025
M.Sc. in Pure Mathematics
GPA: 9.66/10
Thesis: *All modules have flat covers.*
- iii. **National and Kapodistrian University of Athens (NKUA), Greece** 2018 – 2022
B.Sc. in Mathematics (Pure Mathematics Direction)
Grade: 9.46/10

2 Honors and Awards

- **McFarlan Scholarship**, University of Missouri 2025
Awarded for academic excellence upon admission to the Ph.D. program.
- **Graduate Teaching Assistantship**, University of Missouri 2025–
Competitive appointment including tuition support and stipend.

3 Research Interests

Currently, my research focuses on the representation theory of quivers and the application of homological algebra to develop mathematical frameworks and computational pipelines for signal processing, with potential applications in deep learning and data analysis. I am conducting this research under the supervision of Professor Calin Chindris at the University of Missouri.

4 Teaching Experience

- i. **University of Missouri** 2026–2027
 - a. **(Fall)** : Leading Instructor, MATH 1400: Calculus for Social and Life Sciences I
 - b. **(Spring)**:
- ii. **University of Missouri** 2025–2026
 - a. **(Fall)** : Teaching Assistant, College Algebra
 - b. **(Spring)** : Teaching Assistant, College Algebra, and Participation in a Microteaching Seminar preparing for teaching the classes :
 - MATH 1400 – Calculus for Social and Life Sciences I
 - MATH 1300 - Finite Mathematics

iii. **University-Level Tutoring (2018–2025)**

Extensive experience in teaching and tutoring undergraduate and postgraduate mathematics courses at NKUA, NTUA, the University of Patras, and Maastricht University, both independently and in collaboration with the tutoring center SupM.

Courses taught include:

- Real Analysis
- Linear Algebra I & II
- Ordinary Differential Equations
- Differential Geometry of Curves and Surfaces
- Abstract Algebra
- Number Theory
- Probability Theory
- Mathematical Analysis (for Mathematicians and Physicists)
- Mathematical Methods in Physics
- Statistics and Probability (Psychology)
- Mathematics for Chemists
- Algebra II (Graduate)

iv. Also taught High School Mathematics continuously (2018–2025).

5 Publications and Educational Material

5.1 Master's Thesis

All modules have flat covers

National and Kapodistrian University of Athens, 2025.

Field: Homological Algebra.

5.2 Books

Co-author (with Eleftherios Markesinis), *Differential Geometry of Curves and Surfaces*, SupM.

5.3 Lecture Notes ($\text{\LaTeX}$)

Prepared complete lecture notes for:

- Linear Algebra II
- Ordinary Differential Equations
- Homological Algebra and Category Theory
- Riemannian Geometry e.t.c.

Also I have developed structured problem collections in: Galois Theory, Ring and Module Theory, Commutative Algebra, Algebraic Topology, Manifold Theory, Representation Theory of Finite Groups, Mathematical Logic, Set Theory, Differential Geometry, and for other courses.

6 Academic Service/Leadership and Volunteering

President, Postgraduate and Ph.D. Student Association, NKUA (2023–2025). Represented graduate students and liaised with faculty. Co-organizer of the Mathematics Club Seminar Series. Founding member of a volunteer initiative supporting undergraduates through shared lecture notes and past examinations.

Technical Skills and Languages

Advanced proficiency in $\text{\LaTeX}$.

Working knowledge of MATLAB and Python.

English (TOEFL 95/120)